Group 42 Progress Step One:
Category Detection of YouTube Videos

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1 Annotation Instructions

For our task of identifying the primary category of a YouTube video, we've established some guidelines to help annotators with our classification task. To start, a quick summary of the task itself. Our group have determined 12 categories of YouTube videos, sourced from YouTube's category list (see here) and our team's own assumptions.

The task is to answer the classification question, **"What is the main category of this YouTube video?"**. You can achieve this by reading both the **title** and **description** in order to determine the most probably category. In other words, each sample should be labeled with only 1 of the 12 categories.
Use the guideline below to support your decision, but ultimately your best judgement as there will be edge cases. Overall guidelines for annotation are as follows:

1. Use both the video title and the description
2. Do NOT google the YouTube video, as the whole point is to determine based on just the title and description, which watching the video will defeat. However, you may google definition of words.
3. If a video title and description don't align, place more emphasize on the context mentioned from the title.
4. Identify the primary intent, think "What is the video mainly trying to do?". For example, a cooking vlog in Paris should be categorized as Food & Cooking instead of Lifestyle. Since the primary content is on the cooking itself and the creator just happened to be in Paris.
5. Do not categorize based on the channel, we should assume all videos are independent and not categorize all videos similarly if it came from the same channel.
6. If a video can be grouped in multiple categories, choose the category that is more specific. Try thinking: "If I'm on YouTube and browsing by category, which category would the video most likely be grouped under?"

Guidelines for each category are listed below, note that they are not definitive.

1. **Food & Cooking**: Recipe, Cooking Technique
2. **Video Games**: Gameplay, Game Reviews, Streaming Highlights

3. **Sports & Games**: Sport Games, Commentary, Board Games, Training Specific to Sport
4. **Music**: Music Videos, Karaoke Videos
5. **Education**: School Coursework, Math, General Knowledge Videos, Documentary
6. **Shopping**: Product Reviews, Shopping Hauls
7. **Vehicles**: Car Shows, Racing
8. **Lifestyle**: Traveling, Day-in-the-life, Vlogs
9. **News**: Politics, Breaking News, Current Events
10. **Health & Fitness**: Gym, Workout Plans, Diet
11. **Business**: Investing, Economy, Entrepreneurship, Finance
12. **Digital Media**: Movies, Animated Shorts, Fan Fiction, Memes

1.1 Annotation Examples

1.1.1 Food & Cooking

Title: Recreating Maangchi's Seafood Kimchi Fried Rice From Taste | Reverse Engineering | Bon Appétit
Description: We challenged Chris Morocco to recreate Maangchi's seafood kimchi fried rice in the Bon Appétit Test Kitchen. The catch? He's doing it blindfolded with only his other senses to guide him. Want Bon Appétit shirts, hats and more? Still haven't subscribed to Bon Appétit on YouTube? Want more Bon Appétit in your life? Subscribe to the main channel.
Reasoning: This video purpose is on recreating a recipe, so the purpose is to teach the audience how to cook it. It's not Lifestyle/Vlog because we are not watching someone cooking for themselves.

1.1.2 Video Games

Title: A Stupid Stupid Man Sells Sharks in Dave the Diver
Description: Dave has dived too deep, and the sharks now plot their revenge Watch all my Dave the Diver videos: This video was recorded live on my Twitch Channel: Subscribe if you'd like to see more! Become a Youtube Member to support me: Follow me on Twitter:
Reasoning: This video is about a popular videogame called Dave the Diver. Although this requires some background knowledge, we can tell it's not about real life just based on the title, where it says "in" Dave the Diver. If this was a vlog, it would list a place or country, instead of a person.

1.1.3 Lifestyle

Title: Reuniting with my Dad
Description: music from these talented artists;
Reasoning: This is a case where the title and description do not align. We see the description mentions music, while the title is about family reunion. We will use the general guideline of placing more emphasize on the video title, therefore this video should be classified as Lifestyle/Vlog.

1.1.4 Music

Title: Nexpo: Extended Play | Video Extras
Description: Hey everyone. Had some requests for full versions of the new intro/old outro songs that I've used so I've consolidated them here for you to jam to. Hope you enjoy! Track 1 (0:00): "Black Heat" (NE Edit) - Ross Bugden Track 2 (3:57): "Kooler" - ZachJ
Reasoning: Explicit mention of tracks and songs.

1.1.5 Shopping

Title: This Keyboard is BUDGET.
Description: (This keyboard was sent to me for free by Wobkey - I'll have a vid on it this Saturday!) #keyboards #mechanicalkeyboards #mechanicalkeyboard #customkeyboard #keycaps
Reasoning: Although not specifically mentioning "review" or "shop", we can imply from the title that this video is reviewing a keyboard. It won't be lifestyle/vlog because there is no mention of using it for more than a single session, which further implies this is a review.

1.1.6 Sports & Games

Title: MAGNUS vs DING!!!!!!!!!!!
Description: BONUS CONTENT HERE:

Get My Chess Courses: Get my best-selling chess book:

My book in the UK and Europe: Mein Buch auf Deutsch: Mi libro en Espanol: Start Playing Chess FOR FREE: Enjoy my videos? Donate Here :

Email me your games: gothamletters@gmail.com

Reasoning: Chess is a physical board game, so it will belong in Sports & Games, as opposed to video games. However, if this was a chess videogame, we would categorize it as Video Games.

1.1.7 Education

Title: Bayes theorem, the geometry of changing beliefs
Description: Perhaps the most important formula in probability. Help fund future projects: An equally valuable form of support is to simply share some of the videos. Special thanks to these supporters: Home page:

The quick proof:

Interactive made by Reddit user Thoggalluth:

The study with Steve:

Reasoning: This video is educational itself, and not on another category, so we will categorize as Education.

1.1.8 Vehicles

Title: How to PROPERLY Install a Quick Release Steering Wheel (with Working Horn)
Description: Learn how to change your steering wheel so you can remove your old steering wheel and replace it with a quick release steering wheel. In this video I show you how to remove the airbag and steering wheel and install a Momo steering hub, an NRG quick release hub and a racing steering wheel with push buttons for the horn and radio. Thanks @ThatDudeinBlue for sharing your dangerous steering wheel footage! NEW ChrisFix swag: Extendable Ratchet Set:
Reasoning: Even though this video is educational, the primary topic of education is on cars.

1.1.9 News

Title: Contempt for Trump ft. Liz Dye
Description: "That's a lot of contempt Use code LEGALEAGLE50 to get 50% OFF first box and free wellness shots for life with any active subscription: https://legaleagle.link/factor , Need a great lawyer? Welcome back to LegalEagle. The most avian legal analysis on the internets. Watch my next video early & ad-free on Nebula! Suits by Indochino!

GOT A VIDEO IDEA? TELL ME!

Reasoning: Although this video is educating the audience, it's topic is on US politics and covering recent events.

1.1.10 Health & Fitness

Title: The Bodybuilder Who Drinks Chicken Smoothies
Description: Bodybuilder Rene Campbell eats the same high-calorie meal every three hours, seven days a week: a blended chicken shake.

This smoothie blends chicken, rice, cabbage and eggs for one "delicious" (?) protein-heavy shake.

Could you keep this drink down?

#bodybuilder #gymtok #girlwithmuscle #diet #gym #fitness

Subscribe to MUNCHIES here:

Reasoning: Even if the format is somewhat like a documentary, the purpose of the documentary is on a bodybuilder and what they eat. It won't be Cooking/Food since the purpose isn't to promote the meal they make, or showcasing a recipe for their audience to follow.

1.1.11 Business

Title: Worldcoin, the sh*tcoin for humanity just launched
Description: Worldcoin is a cryptocurrency that uses biometric iris data to verify people before making financial transactions. It is a controversial project co-founded by Sam Altman of OpenAI. Some call it a dystopian scam, while others believe it will make the global economy more equitable.

#tech #crypto #thecodereport #ai

Chat with Me on Discord

Resources

Worldcoin Worldcoin raises 115M

Reasoning: This video is about a financial vehicle, cryptocurrency. Even though it covers a recent event, the main message it about world economics and the financial markets.

1.1.12 Digital Media

Title: The Jar Jar storyline we all want #willneff #starwars #movies
Description: Follow me!

TWITCH â†' https://www.twitch.tv/willneff TIKTOK â†' https://www.tiktok.com/@willyneff TWITTER â†' https://twitter.com/TheWillNeff INSTAGRAM â†' https://www.instagram.com/thewillneff/

Find my podcast here â†' @FearAndPodcast

Edited by: https://twitter.com/jookien

Reasoning: This video is about movies and is a fan made digital piece of media.

2 Dataset

The dataset we are using is called youtube-titles by Adam Lucek from Hugging face. It contains 4.9k videos across 50 YouTube Channels sourced from 2024. The dataset is licensed under the MIT License. The following columns can be found in the dataset:

- channel_name: The name of the channel
- channel_url: The URL of the channel
- channel_id: The unique identifier of the channel
- **video_title**: The title of the video
- video_id: The unique identifier of the video
- **video_description**: First 500 characters of the video Description
- prompt: Plausible prompt for generating this title (for LLM training)
- gemma2_9b_it_format: Formatted prompt and title combo for gemma 2 9b instruction tuned model fine tuning

We will only use the `video_title` and `video_description` columns, as bolded in the list above. The dataset is unlabeled for our classification purposes and will be stored as a csv file called `dataset.csv` in our GitHub repository, YoutubeCategories-4NL3, where each row represents one YouTube video.

2.1 Interesting Data Points

We conducted an initial annotation round following our instructions in Section 1. Each of the three annotators spent approximately 1 hour labeling instances using only the `video_title` and the first 500 characters of `video_description`. In this 1-hour estimate, each annotator labeled roughly 250 instances, for about 750 unique labeled videos in total. Following the recommended overlap procedure, we additionally re-annotated 20% of the unique set (150 instances) as a double-annotated overlap set for later agreement calculations.

Quantity	Estimate (1-hour guideline)
# annotators	3
Time spent annotating	$\approx$ 1 hour per annotator
Unique instances labeled	$\approx$ 750 (3×250)
Overlap instances (double-annotated)	$\approx$ 150 (20% of 750)
Total annotation decisions	$\approx$ 900 (750 + 150)
Avg. time per instance	$\approx$ 14.4 seconds (3600 / 250)

Table 1: Estimated annotation volume and speed (update with final counts once annotation is complete).

Below are several edge cases and dataset characteristics we found interesting during annotation, because they create ambiguity even with detailed category guidelines.

1. **Sponsor-heavy descriptions that contain little to no topical signal.**
   Example:
   Title: *I made him cry..*

The corresponding description is dominated by sponsor/affiliate content (e.g., discount codes and links), with minimal information about what happens in the video. This is interesting because our rules require using both title and description, but in practice the description can be uninformative noise. When the title is also vague, the instance becomes difficult to label without channel context (which we explicitly disallow). This suggests models may need robustness to "ad-only" or low-signal descriptions.

2. **Mixed-intent videos that sit between multiple categories (boundary ambiguity).**
   Example:
   Title: *Eating McDonald's Happy Meals until I get EVERY POKEMON CARD!!*
   This single instance plausibly overlaps multiple categories: it involves *food/eating*, a *collecting challenge* (which can resemble Lifestyle/entertainment), and *Pokémon* (which can cue Video Games or Digital Media depending on interpretation). This is interesting because our category list does not include a generic "Entertainment/Challenges" label, so annotators must map these videos to the *closest primary intent* category. These boundary cases are likely to be a major source of annotator disagreement.

3. **Non-English titles and mixed-script text that can still strongly signal a category.**
   Example:
   Title: *å¼·é¢¨ã‚ªãƒ¼ãƒ«ãƒãƒƒã‚¯[ãƒ™ãƒ¼ã‚¹]*
   Description: *Yukopi - å¼·é¢¨ã‚ªãƒ¼ãƒ«ãƒãƒƒã‚¯(feat. æŒæ„›ãƒ¦ã‚) but it's on BASS ...*
   This instance is interesting because the title appears garbled (likely due to text encoding issues), and it contains non-English characters and mixed scripts. Despite this, the description includes clear musical cues (artist name, track-like formatting, "feat.", and "on BASS"), making **Music** the most plausible label. This suggests that models and annotators may need to rely on sparse but high-signal keywords (e.g., "feat.", instrument mentions) and be robust to multilingual or corrupted text when determining categories.

4. **Systematic text noise in scraped metadata (URLs, mojibake, and symbols) that affects pre-processing.** Beyond individual multilingual examples, we observed that many descriptions include heavy URL blocks, affiliate text, and occasional mojibake/encoding artifacts (e.g., corrupted arrows or emoji sequences). This is interesting because it can reduce the effective semantic signal available for classification and can distort tokenization, causing models to overweight frequent non-content tokens (URLs, codes, repeated symbols). This suggests we may want to normalize Unicode, standardize or remove URLs, and apply consistent truncation or filtering rules for descriptions in later experiments.

2.2 Annotation Agreement

To quantify label consistency on the double-annotated overlap set, we computed inter-annotator agreement on the subset of rows where both annotators provided a category label. In the annotated region of 751 instances, 152 instances were double-labeled (20.24% overlap coverage).

Agreement metrics. Because our labels are nominal (unordered) categories, we report both **Cohen's** κ (pairwise agreement corrected for chance) and **Krippendorff's** α (nominal), which naturally supports missing annotations and is appropriate when not all items are annotated by the same number of coders. On the 152 overlapping instances, the **observed agreement** was $P_o = 0.6776$, with **expected agreement** $P_e = 0.1542$, yielding **Cohen's** $\kappa = 0.6189$. Using Krippendorff's α (nominal) on the same overlap set (items with ≥ 2 ratings), we obtained $\alpha = 0.6183$ (with $D_o = 0.3224$, $D_e = 0.8446$). Overall, these values indicate substantial agreement for a title + truncated-description classification setting, suggesting the guidelines are generally interpretable and consistent across annotators.

Segmented overlap analysis. Our annotation process used a "downstream" workflow with annotators labeling different sections of the dataset and overlapping small windows. We computed **pairwise κ separately for each overlap**. We observe consistently moderate-to-very-strong agreement across segments:

- **Segment 1 (rows 0–50,** $n = 51$**):** $P_o = 0.5882$, $P_e = 0.1288$, $\kappa = 0.5274$.
- **Segment 2 (rows 250–300,** $n = 51$**):** $P_o = 0.9020$, $P_e = 0.1726$, $\kappa = 0.8815$.
- **Segment 3 (rows 500–549,** $n = 50$**):** $P_o = 0.5400$, $P_e = 0.1776$, $\kappa = 0.4407$.

Segment 2 shows near-perfect agreement, while Segments 1 and 3 remain moderate. This variability likely reflects differences in example difficulty and boundary ambiguity across windows (e.g., some regions contain more "mixed-intent" or low-signal descriptions than others), rather than a systematic collapse of the shared labeling scheme.

Where disagreements occur. Out of 152 overlap items, there were 49 disagreements. The confusion matrix and sampled mismatches suggest that most inconsistencies arise from **boundary ambiguity** between semantically adjacent categories, rather than random labeling noise. The most frequent and salient confusions include:

- **Lifestyle vs. Education** (the most common mismatch), typically for narrative or experience-driven videos that also contain explanatory, instructional, or "behind-the-scenes" content.
- **Education vs. Video Games / Sports & Games**, especially for puzzle/strategy/game-related content framed as learning, problem-solving, or development (e.g., "coding adventure" or game AI projects).
- **Digital Media vs. Lifestyle / Education**, for short-form edits, commentary, or entertainment clips where the title provides weak topical signal and the description is sparse or template-like.
- **Food & Cooking vs. Shopping / Education**, for cases that resemble product-style series (e.g., tasting or reviewing items) or food history/explanation rather than recipe/technique.

Implications. Overall agreement is strong ($\kappa \approx 0.62$, $\alpha \approx 0.62$), suggesting the dataset is reasonably reliable for downstream modeling. However, the remaining disagreements are concentrated in a few predictable boundary pairs (particularly *Lifestyle* vs. *Education*, and *Education* vs. game-related categories). In future iterations, we plan to add targeted tie-break examples to reduce ambiguity in these regions.

3 Discussion Questions

1. **What did you learn about the task from doing the annotation?**
   We learned that a seemingly simple "title + short description" categorization task quickly becomes a primary-intent reasoning problem. Many titles are vague, and the first 500 characters of the description are often dominated by sponsor templates rather than topical content. As a result, annotators must decide what information is relevant signal versus noise. We also found that the "specific over general" guideline is essential for consistency (e.g., car repair tutorials should be categorized as *Vehicles* rather than *Education*). When descriptions are low-signal, titles tend to be the most stable cue.

2. **What challenges do you expect models to face when learning from your data?**
   We expect models to struggle with low-signal descriptions dominated by URLs, discount codes, affiliate blocks, and social links, as well as vague titles that provide little topical context. Boundary cases

will also be difficult because many instances plausibly fit multiple categories (for example, food challenges can resemble Food & Cooking, Lifestyle, or even Shopping when products are involved). In addition, models may learn spurious correlations from non-topical cues such as `#AD`, repeated brand names, or incidental "music by" credits. Since channel identity is disallowed, models cannot rely on creator/channel priors to resolve ambiguous cases, which may further increase confusion.

3. **What surprising things did you observe in the data?**
   The most surprising observation was how frequently descriptions contain little to no topical signal due to sponsorship blocks and standardized templates. We also encountered systematic text noise such as mojibake and mixed scripts, where the title may appear corrupted while the description still contains usable hints. Another notable surprise was how strongly a few keywords can bias category judgments (e.g., `feat.` suggesting *Music* or `install/steering wheel` suggesting *Vehicles*), even when those cues are sometimes incidental metadata rather than the primary content.

4. **Which features do you expect to be useful?**
   We expect high-signal lexical cues in the title to be most useful, including game names, "recipe", "how to", political figures, and product terms such as "review" or "unboxing". Domain-specific verbs (e.g., `install`, `workout`, `investing`) should help separate categories that all appear educational on the surface. For descriptions, keyword extraction after down-weighting or removing URLs and sponsor blocks should improve topical signal, and named entities (sports teams, politicians, game titles, product brands) can provide strong evidence for the correct category. Lightweight patterns such as tracklists and timestamped song lists may also be useful for identifying *Music* when they reflect the primary intent.

5. **Describe the mental model you came up with to do the task at a high level. What process did you use?**
   We used a two-stage "intent then tie-break" workflow. First, we inferred the primary intent by anchoring on the title and using the description to confirm or disambiguate when it added genuine topical information. Second, when multiple categories seemed plausible, we applied tie-break rules by choosing the more specific category, avoiding channel-based assumptions, and treating sponsor/link-heavy text as non-topical noise. In practice, this meant scanning titles for high-signal tokens, skimming descriptions for topic-confirming terms while ignoring ads, and resolving borderline cases by imagining where the video would most likely appear when browsing YouTube by category.

6. **Was there anything unclear about the instructions? How would you recommend modifying them in the future to make them more helpful?**
   The instructions were mostly clear, but we recommend explicitly stating that sponsor-heavy description blocks and URL dumps should be ignored unless they add genuine topical information. We also recommend adding more guidance for common edge cases, such as Food versus Lifestyle versus Shopping, Gaming versus Digital Media, and News versus Education, either through additional examples or a short tie-break decision guide. Finally, it would help to state clearly that "music by" credits alone do not imply the *Music* category, whereas tracklists and song-like formatting are stronger evidence of music as the primary content.